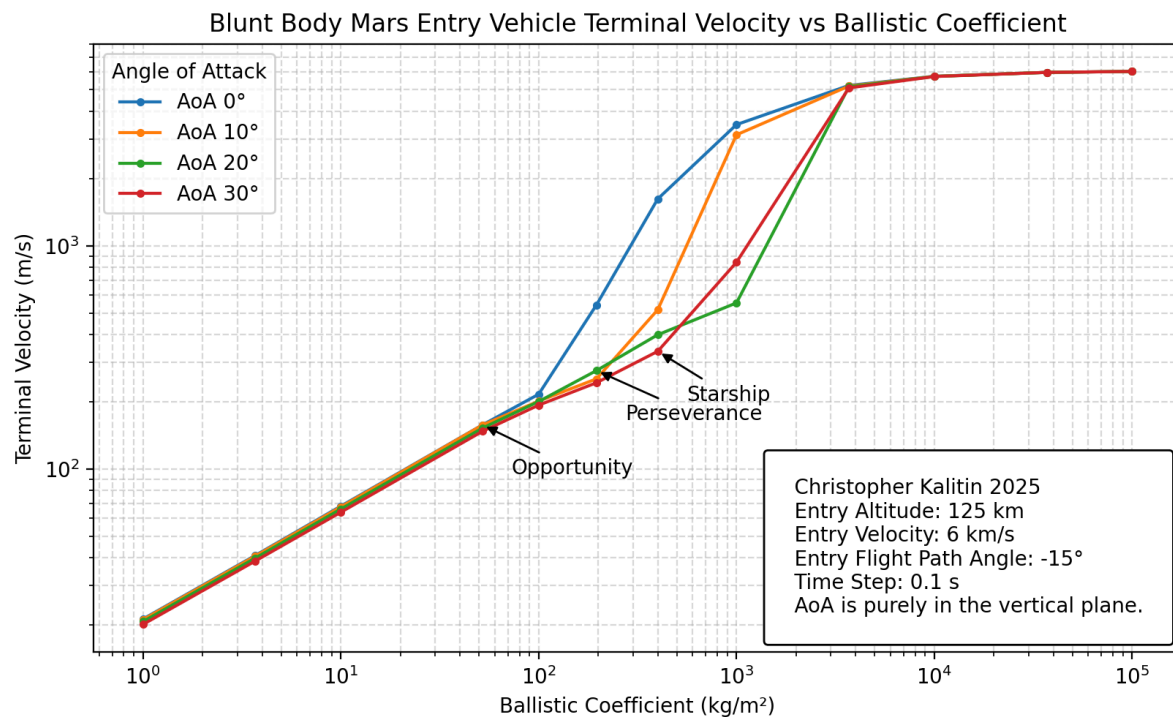


Modelling Terminal Velocity Vs. Ballistic Coefficient For Blunt Body Mars Entry Vehicles

May 13, 2025 - 241 words

Christopher Kalitin Blog (<https://ckalitin.github.io/posts/martian-ballistic-coefficient/>)



Log chart of Terminal Velocity vs. Ballistic Coefficient. If you want an efficient Mars Lander, better be on the left side of the mountain.

Notes

- Problem statement: To land on Mars you need a high ballistic coefficient
- How to land on Mars: Explain the phases of flight (illustrations and charts, explain the charts)
- Model summary (link other blog post)
- Chart summary (general trend and edge cases)

Ballistic Coefficient Points of Interest:

- The first divergence at $BC=100$ occurs because $AoA=0$ no longer reaches terminal velocity. (it's not falling straight down).
- At $BC=196$ (Perseverance) $AoA=20$ is above $AoA=10$ because it's falling straight down after a skip.

- At $BC=1000$, The great divergence occurs because at $AoA=10$ you get a collision and no skip. No skip!
- At $BC=1000$, $AoA=30$ is above $AoA=20$ because it again is falling straight down after a skip.

Structure:

- Introduction
- Companion Post
- Overview of Mars landing
- Max atmospheric deceleration
- Hence, min ballistic coefficient
- Casey Handmer / Starship / Blunt Body examples
- Model:
- First result (in the limit to zero, 0 vel - limit to inf, entry velocity)
- Then, Perseverance test and I realize AoA is extremely important
- AoA Chart with divergence at $\sim 1000 BC$
- First divergence at $BC=100$ because no terminal velocity, and higher AoA reaches v_t faster
- Then, Perseverance at $BC=196$ bc too high skip and $BC=1000$ $AoA=30 > AoA=20$